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Green supply chain network design with stochastic demand and carbon price

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Abstract This paper presents a two-stage stochastic programming model to design a green supply chain in a carbon trading environment. The model solves a discrete location problem and determines the optimal material flows and the number of carbon credits/allowances traded. The study contributes to the existing literature by incorporating uncertainty in carbon price and product demand. The proposed model is applied to a real world case study and the numerical results are carefully analyzed and interpreted. We find that the supply chain configuration can be highly sensitive to the probability distribution of the carbon credit price. More importantly, we observe that carbon price and budget availability for supply chain reconfiguration can both have a positive but nonlinear relationship with greening of the supply chain.

Keywords Supply chain network design · Carbon trading · Environmental regulatory scheme · Green · Stochastic · Case study

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1 Introduction

Strategic supply chain planning decisions involve exploring an optimal network configuration for efficient and effective procurement, production and distribution of material and products from upstream suppliers to the end-users downstream supply chain. This typically includes determining the number, location, capacity, and technology of manufacturing and storage facilities. The strategic configuration of a supply chain has significant influences on tactical and operational planning decisions and hence holds a long-lasting impact on the overall supply chain performance. Substantial capital investments required during the supply chain network design (SCND) phase adds to the significance of these decisions (Li 2013; Santoso et al. 2005).

Traditional SCND approaches have focused on cost-based performance metrics (Dehghanian and Mansour 2009) while the recent regulatory mandates and tighter environmental codes of conduct on the federal, state, and local levels require that organizations expand their set of objectives beyond economic metrics when designing or reconfirming their supply chains (Fahimnia et al. 2015b). The design of eco-efficient supply chains has been recognized as an important step towards the development and management of greener supply chains (Azevedo et al. 2011; Elhedhli and Merrick 2012; Fahimnia et al. 2015c).

Carbon pricing and carbon trading mechanisms are two popular environmental regulatory policy schemes and have been widely practiced in different nations (Zakeri et al. 2015). Carbon pricing aims to control emissions through taxation, in which each greenhouse gas emitter is charged a tax proportional to the size of the emissions generated. This approach is designed to encourage companies to reduce their emissions using methods and technologies whose managerial and implementation costs are less than the tax amount. The primary challenge with carbon pricing is to set the carbon tax to maximize emissions reduction and minimize economic impact (Varsei et al. 2014). The price is usually set for every ton of carbon emitted. Although the charges are relatively stable in a carbon pricing scheme, periodic variations may be needed to achieve certain emissions reduction goals.

In a carbon trading scheme (also known as a cap-and-trade mechanism), a limited number of tradable emissions permits/allowances, the cap, is created for distribution among the players in an economy. Emissions allowances are tradable in a way that companies generating more emissions than the allocated allowances either receive significant fines or purchase emissions allowances off the market from those who generate fewer than the allowed emissions. The scheme creates both pressures (significant fines for over-polluting) and incentive (in the form of a financial reward for selling surplus allowances) to encourage effective environmental management. The two primary challenges with this scheme are determining the allocation of the initial allowances to each company and pricing of tradable allowances in an open market (Zakeri et al. 2015). The price of emissions in this policy is heavily reliant on the supply and demand for allowances in the open market, and consequently subject to a great deal of uncertainty.

Thus, a primary source of uncertainty in both of these regulatory schemes, especially a cap-and-trade environment, is the pricing of carbon or allowances. This introduces an inherent uncertainty factor in supply chain design and planning, which adds to the regular complexities of today's supply chains facing frequent supply and demand interruptions. The latter has been a research focus in several SCND and optimization modeling efforts (Simangunsong et al. 2012). But modeling efforts focusing on uncertainties in carbon/allowance pricing are virtually non-existent. This is the area we wish to contribute to. We study a SCND problem for an organization operating in Australia where the government has been executing carbon

reduction regulatory schemes (currently a carbon tax with a view to convert to a carbon trading scheme in 2015). A two-stage stochastic SCND model is developed in which product demand and carbon credit price parameters are expressed stochastically. Decisions are made at two stages. Stage 1 determines the location, capacity and production technology investments for all supply chain facilities prior to realization of the precise demands and carbon price, and production allocation and distribution planning decisions are made in stage 2, when the demand and carbon prices are realized. The utility of the proposed stochastic model is then investigated using real data from the case company that motivated this research.

The remainder of this paper is organized as follows. Section 2 presents a brief review of the related published models and highlights the contributions of this research that positions our study in the existing literature. The problem definition and the two-stage stochastic model are presented in Sect. 3. Section 4 applies the model to a real world case problem, which arrives at important practical implications and managerial insights. Section 5 presents a summary of the study and results, model and study limitations, and guidance for future research.

2 Literature review

The most recent review of published facility location models indicate that over 82 % of the previous modeling efforts assume deterministic demand and cost parameters (Melo et al. 2009). Limited SCND studies exist that considers uncertainty in critical input parameters, primarily due to the difficulty in solving the resulting complex stochastic models. This appears to be a more recent research trend in the field, especially a focus on demand uncertainty (Bidhandi and Yusuff 2011). The published models can be classified based on the stochastic parameters considered, supply chain scale (i.e. number of tiers), multiplicity of the planning period (i.e. single or multiple periods), product flow considerations (i.e. forward and/or backward flows), type of objective functions (e.g. cost minimization and profit/income/service level maximization) and solution approaches (Baghalian et al. 2013). For example, Sabri and Beamon (2000) studied uncertainty in production, delivery and demand data. Santoso et al. (2005) expressed processing/transportation costs, demands, supplies and capacities as stochastic parameters. Bidhandi and Yusuff (2011) present a two-stage stochastic programming considering uncertainties in operational costs, customer demand and capacity of facilities.

More recently, Baghalian et al. (2013) study the design of a multi-product supply chain considering demand and supply uncertainties. Soleimani et al. (2013) presents a two-stage stochastic programming model to design and plan a closed-loop supply chain with stochastic demand and price of products. Petridis (2013) designs a supply chain network in which demand is normally distributed. Cardoso et al. (2013) develop a multi-stage stochastic mixed integer linear programming model to design and plan a closed-loop supply chain considering uncertainty in product demand.

There are also deterministic SCND modeling efforts that consider environmental performance. For example, Hugo and Pistikopoulos (2005) use a life cycle analysis (LCA) approach in a multi-objective SCND model. Ramudhin et al. (2008) introduce a mixed integer program for a carbon-market-sensitive supply chain problem, investigating the impact of transportation, subcontracting, and production operations on the design of a green supply chain. Bojarski et al. (2009) study a supply chain planning and design problem in which IMPACT 2002+ is used to evaluate the environmental impact assessment of the supply chain. IMPACT 2002+ categorizes environmental damages into four classes of human health, eco-

system quality, climate change and resources and generates a single normalized indicator to estimate the overall environmental impact. Diabat and Simchi-Levi (2009) present a carbon capped supply chain where the throughput and storage capacity of manufacturing site and distribution centers are considered as decision variables. Chaabane et al. (2011) present a methodology for sustainable supply chain design in an emissions trading environment in which logistics costs and carbon emissions are explicitly modeled. Abdallah et al. (2012) present an optimization model for green procurement and supply chain management that minimizes traditional supply chain location and transportation costs, as well as carbon emissions costs through carbon market trading. Chaabane et al. (2012) develop a supply chain design model with reverse logistics using a LCA approach to deal with emissions trading concerns. Devikaa et al. (2014) present a mixed integer programming model for designing a sustainable closed-loop supply chain considering environmental and social impacts.

Greening and emissions reduction initiatives have also been studied at the tactical and operational planning levels. For example, the effectiveness of carbon tax and emissions trading schemes on the forward operations of the supply chain have been investigated by Fahimnia et al. (2014, 2015a) and Zakeri et al. (2015). Carbon taxing impacts on the forward and reverse operations of a closed-loop supply chain have also been studied by Fahimnia et al. (2013). Obviously, not all supply chain design decisions are entirely incorporated in tactical/operational planning level of analysis.

All of these studies assume that the carbon credit price is a deterministic parameter, which is an unrealistic assumption in a cap-and-trade environment. The study of Guillén-Gosálbez and Grossmann (2009) was among the first to consider uncertainty in environmental parameters. In particular, production and transportation emissions are assumed to be stochastic. The focus of this study is on the planning of a chemical supply chain, modeled as a stochastic mixed-integer nonlinear programming model that simultaneously maximizes the net present value of profit and minimizes the environmental impact for a given probability level. Pishvaee et al. (2012) propose a bi-objective credibility-based fuzzy mathematical programming model to determine the configuration of a green logistics network. Production cost and emissions are expressed using fuzzy parameters.

A SCND study considering uncertainty in carbon credit price is non-existent. This is the focus of our study. We present a stochastic model that not only incorporates uncertainty in carbon price, but it also addresses uncertainty in demand parameters. The proposed two-stage model in this paper determines the location, capacity, and production technology investments for all supply chain facilities (stage 1 decisions) as well as production allocation and distribution quantities (stage 2 decisions). The utility of the proposed stochastic model is investigated in a real world case problem and practical implications and managerial insights are presented from the analysis of numerical results.

3 Problem definition and modeling

The members in the supply chain under consideration include suppliers, manufacturing plants, warehouses and end-users. We aim to develop a two-stage stochastic programming model to design or reconfigure a green supply chain in two stages. *Stage 1* decisions (strategic SCND decisions) include (1) the selection of suppliers to fulfill the raw material requirements for the production of multiple product types in manufacturing plants, (2) determining the location and capacity of manufacturing and storage facilities, and (3) determining the production technology requirements at each plant. Production technology affects the acquisition cost,

per-unit production cost and emissions rate. *Stage 2* decisions (tactical SCND decisions) include determining production quantities and material flow across the supply chain. Different modes of transportation may be available between supply chain participants with various per-unit costs and emissions rates. The supply chain operates under an emissions trading scheme. Carbon credit price and product demand are expressed stochastically to reflect the uncertainty in these key input parameters. A finite number of scenarios are considered for product demand and carbon price. Stage 1 decisions are made based on all scenarios and Stage 2 decisions are made upon the realization of demand values and carbon credit price.

The general form of the two-stage stochastic program is as follows:

$$\min z = cx + E_{\boldsymbol{\xi}} Q(x, \boldsymbol{\xi}) \tag{1}$$

$$\text{subject to: } Ax = b \text{ and } x \geq 0 \tag{2}$$

where x notes the stage 1 decision variables, c is the row vector of coefficient of x in objective function of stage 1, A is the matrix of coefficient of variables in constraints set and b is the right hand side column vector of problem in stage 1. Also, $Q(x, \xi) = \min\{q^T y | Wy = h - Tx, y \geq 0\}$, where y denotes the stage 2 decisions, ξ is the vector formed by components of q^T(row vector of coefficient of decision variables in objective function of stage 2), W (coefficient matrix of variables in constraints set of stage 2 problem), h (right hand side column vector of stage 2 problem), and T (coefficient matrix of stage 1 variables in stage 2 problem). E_ξ (the mathematical expectation with respect to $\boldsymbol{\xi}$) corresponds to the expected value of the cost of stage 2 decisions. Boldface notations are used to indicate that these vectors are random and to differentiate them from their actual values after realization.

We assume that the random vector ξ has a finite number of states. Let $k = 1, \ldots, K$ index its possible realizations and p_k be their respective probabilities. Under this assumption, we can now express the extensive form of the two-stage stochastic program in Eqs. 3–6 (Birge and Louveaux 1997).

$$\min c^T x + \sum_{k=1}^{K} p_k q_k^T y_k \tag{3}$$

$$\text{Subject to: } Ax = b, \tag{4}$$

$$T_k x + W y_k = h_k, \quad k = 1, \ldots, K, \tag{5}$$

$$x \geq 0, \quad y_k \geq 0, \quad k = 1, \ldots, K. \tag{6}$$

The supply chain under consideration is illustrated in Fig. 1 using notations defined below.

Indices

i Product index
r Raw material index
n Supplier index
m Manufacturing plant index
h Production technology
u Capacity index in Manufacturing plant
w Warehouse index
v Capacity level in warehouse
k Transport mode index
j End-user index
s Scenario index

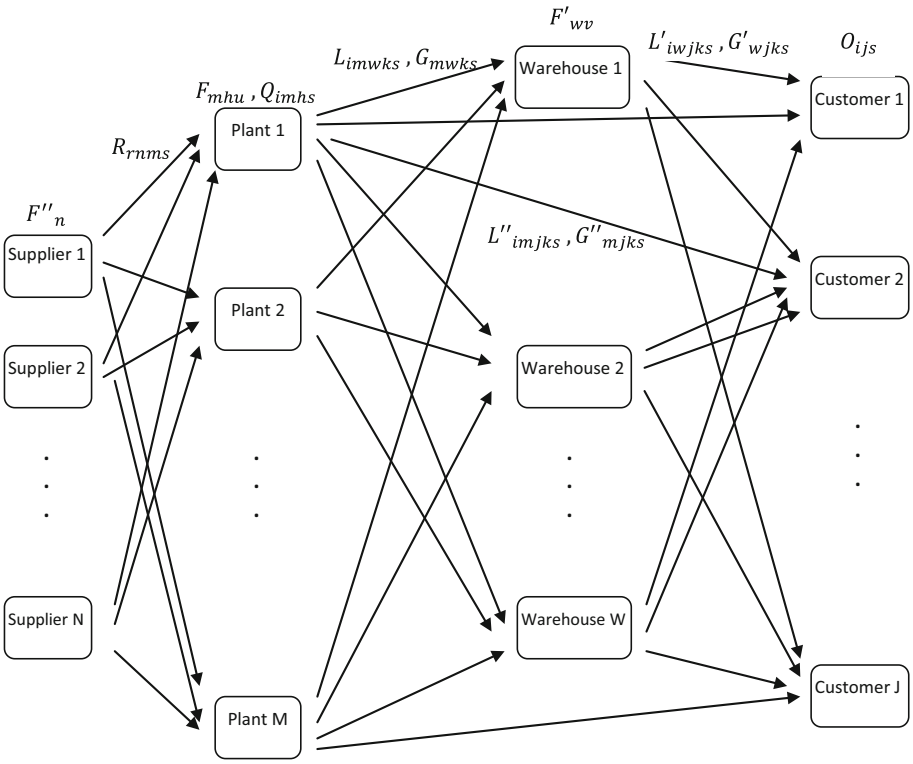


Fig. 1 Decision variables in the supply chain under consideration

Input Parameters

π_s	Market price of carbon credit in scenario s
d_{ijs}	Forecasted demand for product i in end-user j in scenario s
cap	Maximum overall allowed carbon emissions (ton)
f_{mhu}	Fixed costs of establishing plant m with technology h and capacity u
f'_{wv}	Fixed costs of establishing warehouse w with capacity v
f''_n	Fixed costs of selecting supplier n
cm_{imh}	Cost of manufacturing a unit of product i with technology h in plant m
cs_{rnm}	Cost of purchasing a unit of raw material r from supplier n for processing at plant m
ct_{imwk}	Per unit cost of shipment of product i from plant m to warehouse w through transportation mode k
ct'_{iwjk}	Per unit cost of shipment of product i from warehouse w to end-user j through mode k
ct''_{imjk}	Per unit cost of shipment of product i from plant m to end-user j through mode k
ρ_{ih}	Processing time of a unit of product i with technology h (h)
ρ'_i	Volume of a unit of product i (m^3)
α_{ri}	Amount of raw material r required in production of a unit of product i (kg)
c_{mhu}	Production capacity (time) in plant m with technology h and capacity u (h)
c'_{wv}	Holding capacity in warehouse w with capacity level v (m^3)

c''_{rn} Raw material capacity for raw material r in supplier n (kg)
lb_{mwk} Minimum allowed transportation from plant m to warehouse w through mode k (m^3)
lb'_{wjk} Minimum allowed transportation from warehouse w to end-user j through mode k (m^3)
lb''_{mjk} Minimum allowed transportation from plant m to end-user j through mode k (m^3)
ub_{mwk} Maximum allowed transportation from plant m to warehouse w through mode k (m^3)
ub'_{wjk} Maximum allowed transportation from warehouse w to end-user j through mode k (m^3)
ub''_{mjk} Maximum allowed transportation from plant m to end-user j through mode k (m^3)
em_{imh} Estimated carbon emissions to produce a unit of product i with technology h in plant m (ton)
et_{imwk} Estimated carbon emissions for shipping a unit of product i from plant m to warehouse w through mode k (ton)
et'_{iwjk} Estimated carbon emissions for shipping a unit of product i from warehouse w to end-user j through mode k (ton)
et''_{imjk} Estimated carbon emissions for shipping a unit of product i from plant m to end-user j through mode k (ton)
lt_{mh} Conversion parameter of fixed costs of establishing plant m with technology h into annual equivalent
lt'_{w} Conversion parameter of fixed costs of establishing warehouse w into annual equivalent
pc_{ij} Penalty cost for shortage of a unit of product i in end-user j
ps_s Probability of scenario s
b Total budget for establishment of facilities

Continues Decision Variables

Q_{imhs} Amount of product i produced in plant m with technology h in scenario s
R_{rnms} Amount of raw material r shipped from supplier n to plant m in scenario s
L_{imwks} Amount of product i shipped from plant m to warehouse w through k in scenario s
L'_{iwjks} Amount of product i shipped from warehouse w to end-user j through mode k in scenario$s$
L''_{imjks} Amount of product i shipped from plant m to end-user j through mode k in scenario s
O_{ijs} Amount of shortage for product i in end-user j in scenario s
E_s Net number of carbon credits traded in scenario s

Binary Decision Variables

$$\mathrm{F}_{mhu} = \begin{cases} 1, & \text{If manufacturing plant } m \text{ with technology } h \text{ and capacity } u \text{ is established} \\ 0, & \text{Otherwise} \end{cases}$$

$$F'_{wv} = \begin{cases} 1, & \text{If warehouse } w \text{ with capacity level } v \text{ is established} \\ 0, & \text{Otherwise} \end{cases}$$

$$F''_{\mathrm{n}} = \begin{cases} 1, & \text{If supplier } n \text{ is selected} \\ 0, & \text{Otherwise} \end{cases}$$

$$G_{mwks} = \begin{cases} 1, & \text{if there is a flow between plant m and warehouse } w \text{ through mode } k \text{ in scenarios} \\ 0, & \text{Otherwise} \end{cases}$$

$$G'_{wjks} = \begin{cases} 1, & \text{if there is a flow between warehouse } w \text{ and end-user } j \text{ through mode } k \text{ inscenarios} \\ 0, & \text{Otherwise} \end{cases}$$

$$G''_{mjks} = \begin{cases} 1, & \text{if there is a flow between plant } m \text{ and end-user } j \text{ through mode } k \text{ inscenarios} \\ 0, & \text{Otherwise} \end{cases}$$

The model in this section is presented based on the extensive form of the stochastic model shown in Eqs. 3–6. In the proposed stochastic SCND model, the objective function Z in Eq. 7 formulates the overall SCND costs at the strategic and tactical levels. This includes the fixed cost of establishing plants and warehouses, the fixed cost of supplier selection, and the expected cost of tactical decisions.

$$\begin{aligned} Z = & \sum_m \sum_h \sum_u (lt_{mh} f_{mhu})\, F_{mhu} \;+\; \sum_w \sum_v \left(lt'_w f'_{wv}\right) F'_{wv} + \sum_n f''_n F''_n \\ & + \sum_s ps_s \left(\sum_i \sum_m \sum_w \sum_k ct_{imwk} L_{imwks} + \sum_i \sum_w \sum_j \sum_k ct'_{iwjk} L'_{iwjks} \right. \\ & + \sum_i \sum_m \sum_j \sum_k ct''_{imjk} L''_{imjks} + \sum_i \sum_m \sum_h cm_{imh} Q_{imhs} \\ & \left. + \sum_r \sum_n \sum_m cs_{rnm} \mathrm{R}_{\mathrm{rnms}} + \sum_i \sum_j pc_{ij} O_{ijs} + \pi_s E_s \right) \end{aligned} \tag{7}$$

The goal of the proposed model is to minimize the value of the objective function Z in Eq. 7. The objective function formulates the cost of establishing manufacturing plants and warehouses (terms 1 and 2), the fixed cost of supplier selection (term 3), expected values of transportation cost between supply chain participants (terms 4–6), production and raw material purchasing costs (terms 7 and 8), penalty/shortage cost (term 9), and the net cost of carbon credits (term 10). The objective function is subject to the following constraints:

$$\sum_m \sum_h \sum_u f_{mhu} F_{mhu} + \sum_w \sum_v f'_{wv} F'_{wv} \le b \tag{8}$$

$$\sum_h \sum_u F_{mhu} \le 1 \quad \forall m \tag{9}$$

$$\sum_v F'_{wv} \le 1 \quad \forall w \tag{10}$$

$$\begin{aligned} & \sum_i \sum_m \sum_w \sum_k et_{imwk} L_{imwks} + \sum_i \sum_w \sum_j \sum_k et'_{iwjk} L'_{iwjks} \\ & + \sum_i \sum_m \sum_j \sum_k et''_{imjk} L''_{imjks} + \sum_i \sum_m \sum_h em_{imh} Q_{imhs} - cap = E_s \quad \forall s \end{aligned} \tag{11}$$

$$\sum_i \rho_{ih} Q_{imhs} \le \sum_u c_{mhu} F_{mhu} \quad \forall m, h, s \tag{12}$$

$$\sum_i \sum_m \sum_k \rho'_i L_{imwks} \le \sum_v c'_{wv} F'_{wv} \quad \forall w, s \tag{13}$$

$$\sum_{m} \mathrm{R}_{\mathrm{rnms}} \leq \mathrm{c}''_{\mathrm{rn}} \mathrm{F}''_{\mathrm{n}} \quad \forall r, n, s \tag{14}$$

$$\sum_{i} \sum_{h} \alpha_{ri} Q_{imhs} = \sum_{n} \mathrm{R}_{\mathrm{rnms}} \quad \forall r, m, s \tag{15}$$

$$\sum_{h} Q_{imhs} = \sum_{w} \sum_{k} L_{imwks} + \sum_{j} \sum_{k} L''_{imjks} \quad \forall i, m, s \tag{16}$$

$$\sum_{m} \sum_{k} L_{imwks} = \sum_{j} \sum_{k} L'_{iwjks} \quad \forall i, w, s \tag{17}$$

$$\sum_{w} \sum_{k} L'_{iwjks} + \sum_{m} \sum_{k} L''_{imjks} + O_{ijs} = d_{ijs} \quad \forall i, j, s \tag{18}$$

$$lb_{mwk} G_{mwks} \leq \sum_{i} \rho'_i L_{imwks} \leq ub_{mwk} G_{mwks} \quad \forall m, w, k, s \tag{19}$$

$$lb'_{wjk} G'_{wjks} \leq \sum_{i} \rho'_i L'_{iwjks} \leq ub'_{wjk} G'_{wjks} \quad \forall w, j, k, s \tag{20}$$

$$lb''_{mjk} G''_{mjks} \leq \sum_{i} \rho'_i L''_{imjks} \leq ub''_{mjk} G'_{mjks} \quad \forall m, j, k, s \tag{21}$$

$$F_{mhu}, F'_{wu}, F''_{n}, G_{mwks}, G'_{wjks}, G''_{mjks} \in \{0, 1\} \quad \forall m, h, u, w, n, k, s, j \tag{22}$$

$$Q_{imhs}, R_{rnms}, L_{imwks}, L'_{iwjks}, L''_{imjks} \geq 0 \quad \forall i, m, h, s, r, n, m, s, w, k, j \tag{23}$$

$$E_s \text{ is URS} \quad \forall s \tag{24}$$

Constraint (8) enforces the budget limitation for the establishment of manufacturing plants and warehouses. Constraints (9) and (10) ensure that no more than one facility can be established in each candidate location. Constraint (11) presents the carbon credit limitation at each scenario by subtracting the production and transport emissions from the regulated cap. Constraints (12) and (13) enforce the capacity limitation of the manufacturing plants and warehouses. Constraint (14) formulates the limitation in raw material provision by suppliers. Constraint (15) guarantees the fulfillment of raw material requirement in the manufacturing plants. Constraints (16), (17) and (18) enforce the flow balance in manufacturing plant, warehouse and end-user locations, respectively. Constraints (19), (20) and (21) impose the product flow limitations between different supply chain participants. Constraints (22)–(24) enforce restrictions on decision variables.

4 Model implementation: a case study

The case company, referred to as ABC, is engaged in the production, assembly and distribution of a broad range of office furniture in Australia. This case study concerns the sustainable strategic design of ABC's supply chain network that provides three popular product types (product families) to end-users located in five different states of New South Wales, Victoria, Queensland, South Australia and Western Australia. Each state has a potential location for establishing manufacturing plants and warehouses (there are hence five potential locations for plants and five for warehouses). Three suppliers in Queensland, South Australia and Western Australia can provide the required raw material and components to these manufacturing plants. A manufacturing plant can be built in small, medium or large capacity; while a warehouse can be either small or large in size. Three production technologies are available

at each manufacturing plant. A greener technology incurs higher acquisition cost, but may have lower production cost. Up to three transport modes with various costs and emissions rates are available at each route. ABC wishes to impose a budget constraint of \$30M for the reconfiguration of its existing supply chain.

Australia introduced a two-stage set of regulatory environmental policy mechanisms in 2012. In the first stage, initiated in July 2012, organizations respond to a carbon taxation (pricing) scheme. A ton of carbon pollution is priced at \$23 in 2012 rising by 5 % per year. At the second stage, after 3 years of fixed-price period, the scheme is expected to convert into a full emissions trading mechanism in 2015. Our analysis in this section aims to evaluate the ABC's financial and emissions performance in an emissions trading environment. The emissions cap is set equal to 90 % of the overall emissions generated by ABC in 2013. That is, the goal is to reduce the overall generated emissions by 10 % through supply chain restructuring.

We consider a set of demand and carbon price situations to develop a number of scenarios for our stochastic analysis. Two situations are defined for demand of each product type. Given the market performance, demand for a product can be either "optimistic" or "pessimistic" in all end-users locations. With three product types, eight demand scenarios are resulted (2*2*2 = 8). For carbon price, we define three situations: low, medium and high. Using these demand and carbon price situations, for the 3 product types, we have 24 scenarios (8*3 = 24), all with equal probability of occurrence. The model is coded in AMPL modeling language using CPLEX as the core optimization solver.

4.1 Numerical results for deterministic and stochastic models

We first solve the deterministic-equivalent of the two-stage stochastic model presented in Sect. 3. All uncertain parameters are replaced with their expected values. Table 1 summarizes the solutions of the deterministic model for a range of carbon prices. Three different SCND configurations are resulted from carbon prices between \$0 and \$100. The locations of manufacturing plans and warehouses for each configuration are shown in Figs. 2, 3 and 4. Values in these figures denote the capacity of the sites.

Having no carbon regulatory scheme in place (i.e. a carbon price of \$0) results in "configuration 1". We refer to this as the "*base configuration*" with three plants, two of which are equipped with polluting equipment and one with semi-green equipment. Introducing a carbon price of \$20 per ton results in a shift from configuration 1 to 2, which is a greener supply chain configuration. Since transportation is the primary contributor to the carbon emissions (see Table 1), the change in the location and capacity of one plant in configuration 2 ensures reduced inbound transportation costs and emissions. The supply chain tends to buy carbon credits if carbon is priced no more than \$30 per ton (evidenced by the positive numbers in "net carbon cost" column in Table 1). Carbon prices smaller than \$20 per ton have no influence on SCND decisions.

For a carbon price of \$40 and above, the model chooses configuration 3 that requires establishing 4 manufacturing plants and 2 warehouses. An increase in the number of plants allows a reduction in outbound transportation cost and emission, hence the focus is still placed on more efficient and environmentally friendly transport routing rather than greener manufacturing technology. Configuration 3 results in the supply chain using the carbon trading as a source of income (indicated by negative numbers in "net carbon cost" column in Table 1). While configuration 3 is the greenest design amongst the three, it uses less of green manufacturing technologies and only focuses on greening the transportation operations through reduced travel distances (setting up manufacturers closer to

Table 1 Numerical results of the deterministic model for a range of carbon credit prices

Carbon price ($/ton)	Configuration	Fixed cost (M$)	Transportation cost (M$)	Production cost (M$)	Transportation carbon (thousand ton)	Production carbon (thousand ton)	Total carbon (thousand ton)	Net carbon cost ($)	Objective value (M$)
0	1	29.9	15.2	11.7	79.1	28.3	107.4	0	47.89
10	1	29.9	15.2	11.7	79.1	28.3	107.4	123,854	48.02
20	2	28.9	14.3	13	61	34.9	95.9	17,969	48.08
30	2	28.9	14.3	13	61	34.9	95.9	26,953	48.09
40	3	29.8	14.5	13	51.5	34.9	86.3	−347,521	48.09
50	3	29.8	14.5	13	51.5	34.9	86.3	−434,401	48
60	3	29.8	15.3	13	37.5	34.9	72.4	−1,357,883	47.86
70	3	29.8	15.3	13	37.5	34.9	72.4	−1,584,197	47.63
80	3	29.8	15.3	13	37.5	34.9	72.4	−1,810,511	47.41
90	3	29.8	15.3	13	37.5	34.9	72.4	−2,036,824	47.18
100	3	29.8	15.3	13	37.5	34.9	72.4	−2,263,138	46.95

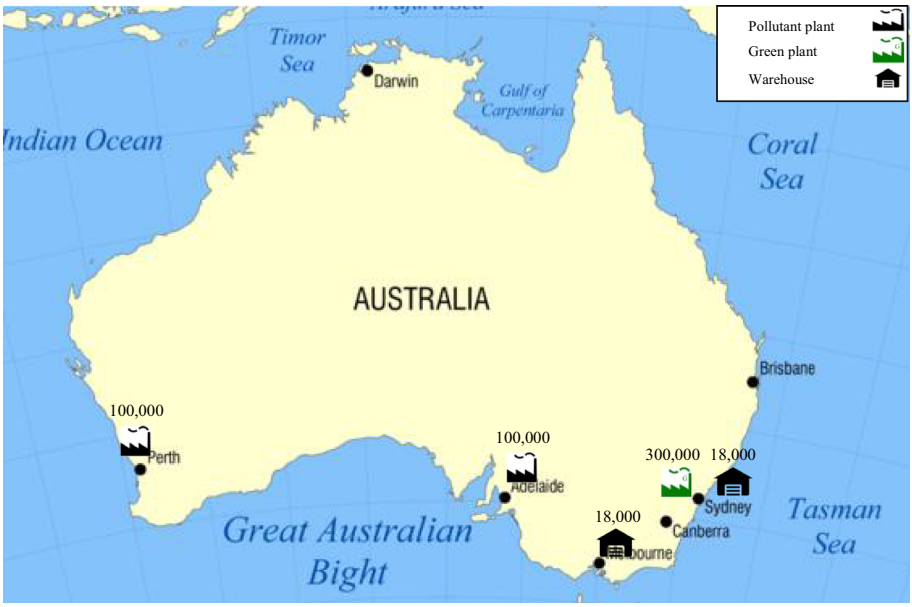


Fig. 2 The location and capacity of manufacturers and warehouses in "Configuration 1" for carbon price of \$0–\$10



Fig. 3 The location and capacity of manufacturers and warehouses in "Configuration 2" for carbon price of \$20–\$30

warehouses and end-users) and the use of greener transport modes. In other words, the model finds transportation initiatives cheaper to implement than the adoption of greener production technologies. Finally, the "Objective Value" column shows that the greener

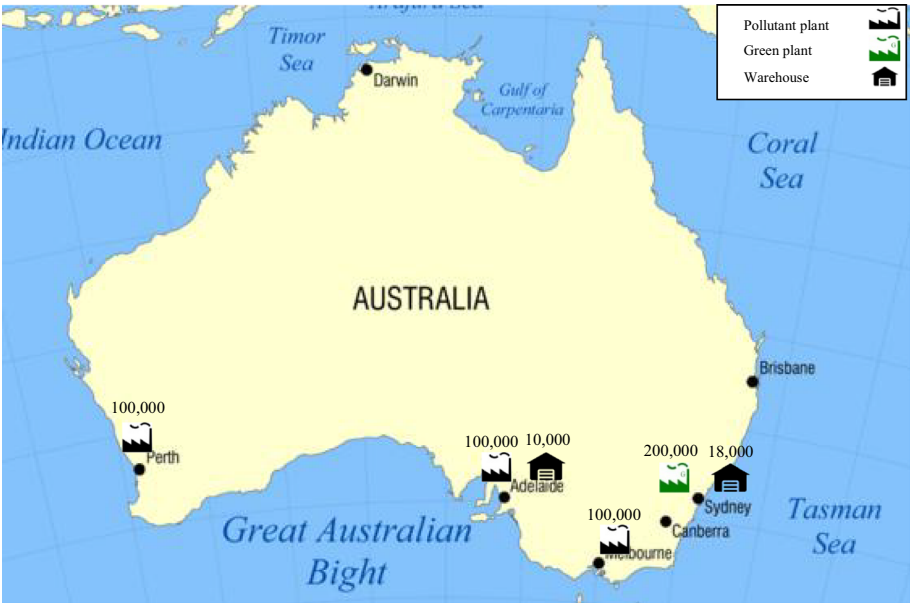


Fig. 4 The location and capacity of manufacturers and warehouses in "Configuration 3" for carbon price of $40–$100

configurations are less expensive to run through selling carbon credits at relatively high prices.

To examine the effect of carbon price uncertainty on SCND decisions, we consider 10 scenario groups for carbon price with identical demand scenarios. Each scenario group constitutes of 24 scenarios as defined earlier in this section, reflecting variations in carbon price and demand status. Table 2 shows the carbon cost variations for each scenario group (see the first four columns). Each row of this table corresponds to a stochastic SCND problem with specific carbon price scenarios. Comparing Tables 1 and 2 (deterministic versus stochastic carbon price) reveals that mid-range carbon prices yield different supply chain configurations. In stochastic scenarios 4, 5 and 6 where the Expected Value of Carbon Price (EVCP) is equal to $40, $50 and $60 respectively, configuration 2 yields the lowest cost, while configuration 3 yields the lowest cost for the deterministic model. In other scenarios (carbon prices of lower than $30 and higher than $70 per ton), both stochastic and deterministic models yield similar configurations.

The numerical results of the stochastic model show two breakpoints where there is a change of supply chain configuration: scenario groups 1–2 change of configuration 1 to 2, and scenario groups 6–7 change of configuration 2 to 3. This result is highly sensitive to the probability distribution of carbon credit price. That is, assuming a slightly different distribution probability function for carbon credit price may result in a different configuration. A cautionary approach for carbon credit prices at these breakpoints ($10–$30 and $50–$80) would be to establish the facilities that are commonly suggested in the two configurations and wait until the realization of actual carbon price for other facility decisions. Similar to the deterministic model, the stochastic model confirms that a concurrent reduction in transportation cost and emissions yields a more efficient transport network (i.e. locating

Table 2 Numerical results of the stochastic model for different carbon credit price scenarios

Scenario groups	Low price ($)	Medium price ($)	High price ($)	EVCP ($)	Configuration	Expected values						Objective value (M$)
						Transportation cost (M$)	Production cost (M$)	Transportation carbon (thousand tons)	Production carbon (thousand tons)	Total carbon (thousand tons)	Net carbon cost ($)	
1	0	10	20	10	1	15.29	11.88	80.31	29.3	109.61	145530	48.3
2	10	20	30	20	2	14.51	12.99	64.74	35.04	99.78	87155	48.38
3	20	30	40	30	2	14.5	13.03	63.6	35.21	98.81	107745	48.42
4	30	40	50	40	2	14.52	13.04	62.52	35.29	97.82	108152	48.46
5	40	50	60	50	2	14.96	13.04	54.48	35.29	89.77	−340780	48.45
6	50	60	70	60	2	15.42	13.02	46.63	35.21	81.83	−866010	48.37
7	60	70	80	70	3	15.59	13.09	38.066	35.53	73.6	−1498193	48.19
8	70	80	90	80	3	15.59	13.09	38.07	35.53	73.6	−1712220	47.98
9	80	90	100	90	3	15.59	13.09	38.07	35.53	73.6	−1926248	47.77
10	90	100	110	100	3	15.59	13.09	38.07	35.53	73.6	−2140276	47.55

the facilities closer to the end-users to minimize the travel distances). On the other hand, reduction in transport emissions together with an increased transportation cost requires the use of greener transportation modes.

4.2 The impact of budget restriction on SCND decisions

To evaluate the impact of budgetary constraints on SCND decisions, we solve the stochastic model for budgets ranging from \$20M to \$45M for three EVCPs of \$10, \$50 and \$100, corresponding to low, medium and high carbon prices. Numerical results are presented in Table 3. It is evident from these results that smaller budgets limit the number of plants and warehouses which results in increased transportation cost and emission. Another observation at more constrained budgets is the use of more polluting machinery in manufacturing plants which requires smaller capital investments, but leads to increased on-going production cost and carbon emissions. Interestingly, even though the initial setup costs (fixed SCND costs) are higher for more constrained budgets, the objective function value (overall supply chain operating costs) are reduced by approximately 19 %.

The difference between supply chain configurations for the two ends of the budget spectrum (i.e. \$20M and \$45M budget constraints) are shown in Figs. 5 and 6. Not surprisingly, greener configurations can be achieved for higher budgets. This observation can provide a policy insight on how and on what basis governments can provide financial incentives during the supply chain design phase for the development of greener supply networks. For example, these results show what additional budgets need to be injected into the supply chain to achieve a given emissions reduction target.

For the range of budget constraints, Fig. 7 illustrates the total carbon emissions at different carbon credit prices. The total emissions are clearly lower at higher EVCPs, but what is interesting is the nonlinear behavior of emissions reduction as the budget increases. On average, every \$5M budget escalation results in about 10 % emissions reduction for all EVCPs. However, this emissions improvement occurs as a faster rate at the initial budget increases. This provides an important policy implication: providing smaller financial incentives to multiple businesses may be more environmentally beneficial when compared to providing larger incentives to a limited number of players (from a global/national/regional sustainability perspective).

4.3 The impact of carbon regulation on supply chain decisions

Next, we evaluate the impact of emissions trading regulations on the strategic and tactical supply chain decisions. We define two cases. *Case 1* is when the supply chain is in the functional stage. In this case, strategic facility location and technology decisions are already made and adjustments can only be made to tactical decisions, such as production allocation and distribution. *Case 2* is when the supply chain is in the *design stage* which involves both strategic location/technology decisions and tactical allocation decisions. These situations are studied in *three environmental regulatory environments*: (1) when there is no emission regulation in operation, (2) when an emission cap is imposed with no emissions trading mechanism, and (3) when a full cap-and-trade scheme (carbon trading mechanism) is practiced. In all experiments, carbon price (if applicable) and demand are considered as stochastic parameters.

For Case 1 (a functional supply chain), it is assumed that the supply chain configuration is determined when no carbon regulatory scheme is in place (i.e. \$0 carbon price in the

Table 3 Numerical results of the stochastic model for various budgetary constraints

Budget constraint (M$)	Fixed establishment cost ($)			Transportation cost ($)			Production cost ($)			Objective function value ($)		
	EVCP = $10	EVCP = $50	EVCP = $100	EVCP = $10	EVCP = $50	EVCP = $100	EVCP = $10	EVCP = $50	EVCP = $100	EVCP = $10	EVCP = $50	EVCP = $100
20	19,497,500	19,689,500	19,689,500	16,855,281	17,998,229	19,429,448	15,432,000	15,432,000	15,432,000	51,817,600	53,627,400	54,015,700
25	24,999,500	24,539,000	24,539,000	15,148,319	14,794,279	15,143,463	14,252,322	15,432,000	15,432,000	49,534,700	49,947,100	49,486,600
30	29,921,250	28,909,500	29,811,250	15,294,016	14,960,763	15,589,195	11,880,581	13,041,594	13,091,682	48,299,400	48,453,500	47,552,900
35	34,609,500	34,609,500	34,609,500	14,423,904	14,910,691	15,783,425	11,129,813	11,130,219	11,129,813	47,425,600	47,115,300	45,726,600
40	38,609,500	38,609,500	38,609,500	13,119,554	13,575,747	14,440,372	11,129,813	11,130,094	11,129,813	46,871,700	46,395,500	44,827,400
45	44,000,500	44,876,000	44,876,000	12,719,808	13,348,341	13,653,417	10,288,000	10,288,000	10,288,000	46,654,300	45,919,600	43,963,800



Fig. 5 The supply chain configuration for a $20M budget availability



Fig. 6 The supply chain configuration for a $45M budget availability

proposed stochastic model). In Sect. 4.1, we saw that the resulting SCND for this setting is configuration 1, as illustrated in Fig. 2. This configuration remains unchanged for situation 1 and the regulatory environment only affects the production and distribution decisions for this

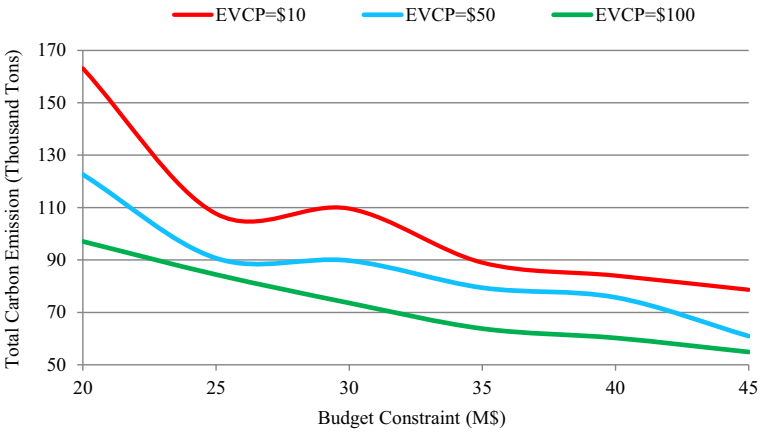


Fig. 7 Supply chain emission performance as a function of budget constraint at different carbon credit prices

fixed configuration. Table 4 summarizes the numerical results for all regulatory environments. Similar to the former stochastic experiments we use three EVCPs of $10, $50 and $100 for regulatory environment 3 in a carbon trading regulatory environment.

Considering the "no emission regulation" situation as the base regulatory environment, the results show that imposing an emissions cap of 95,000 tons results in a 1.89 % increase in the objective function value, while the emissions performance is improved by approximately 15 %. In a carbon trading environment, the objective function values increases by 0.83, 1.38 and 1.50 % for EVCPs of $10, $50 and $100 respectively. Emissions reductions in these cases are equal to 6.71, 11.31 and 12.30 %, respectively. A carbon cap mechanism is therefore more expensive to operate, but achieves larger carbon reductions. An equivalent emissions performance in a cap-and-trade scheme can only be achieved at carbon prices of greater than $100 per ton of emission. From a policy perspective, the most effective scheme depends on emissions reduction goals as well as the economic situation (i.e. the amount of pressure regulations can exert on the industry).

For Case 2 (a supply chain at the design stage), the primary decision is to determine an optimal supply chain configuration. Numerical results for various regulatory schemes are shown in Table 5. Similar to the functional supply chain case, the no-emission-regulation environment results in configuration 1. Considering a carbon cap yields a greener configuration, an approximately 16 % emissions reduction, with a corresponding 1.22 % increase in the objective value. In the presence of a carbon trading scheme, the optimal supply chain configuration may vary depending on the carbon credit price. In the best case scenario, a 32.89 % emissions improvement is observed at an EVCP of $100. This yields a 1.25 % improvement in the objective function value, triggered by the additional revenue from selling emissions allowances.

Figures 8 and 9 illustrate the supply chain cost and emissions performance in the two cases (functional and design stages) under different carbon regulatory mechanisms against the baseline of no-emission-regulation. Based on these comparisons, green initiatives at the design stage can be less costly (Fig. 8) and more environmentally beneficial than those at the functional stage (Fig. 9).

Table 4 Numerical results of Case 1: supply chain at the functional stage

Regulatory environment	Configuration	Establishing cost ($)	Production cost ($)	Production emission (ton)	Transportation cost ($)	Transportation emission (ton)	Objective function ($)	Changes in objective function value (%)	Total emissions (ton)	Emissions improvement compared to the base scenario (%)
No emission regulation	1	29,921,250	11,880,581	29,299	15,293,256	80,362	48,153,100	–	109,660	–
Carbon cap only	1	29,921,250	11,888,688	29,027	16,193,855	63,873	49,061,750	1.89	92,900	15.28
Emissions trading with EVCP = $10	1	29,921,250	11,883,283	29,208	15,688,183	73,095	48,550,750	0.83	102,303	6.71
Emissions trading with EVCP = $50	1	29,921,250	11,889,333	29,030	15,948,394	68,233	48,816,951	1.38	97,263	11.31
Emissions trading with EVCP = $100	1	29,921,250	11,888,688	29,027	16,005,624	67,141	48,873,552	1.50	96,167	12.30

Table 5 Numerical results of Case 2: supply chain at the design stage

Regulatory environment	Configuration	Establishing cost ($)	Production cost ($)	Production emission (ton)	Transportation cost ($)	Transportation emission (ton)	Objective function ($)	Changes in objective function value (%)	Total emissions (ton)	Emissions improvement compared to the base scenario (%)
No emission regulation	1	29,921,250	11,880,581	29,299	15,293,256	80,362	48,153,100	–	109,660	–
Carbon cap only	2	28,909,500	12,951,619	34,858	14,995,995	56,805	48,739,500	1.22	91,663	16.41
Emissions trading with EVCP = $10	1	29,921,250	11,880,581	29,299	15,294,016	80,308	48,299,400	0.30	109,607	0.05
Emissions trading with EVCP = $50	2	28,909,500	13,041,594	35,290	14,960,763	54,479	48,453,500	0.62	89,770	18.14
Emissions trading with EVCP = $100	3	29,811,250	13,091,682	35,530	15,589,195	38,067	47,552,900	−1.25	73,597	32.89

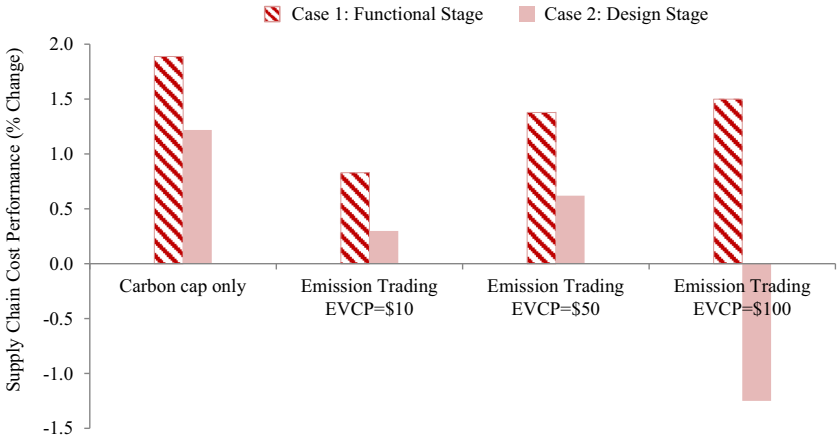


Fig. 8 Supply chain cost performance at the functional and design stages under different carbon regulatory environments

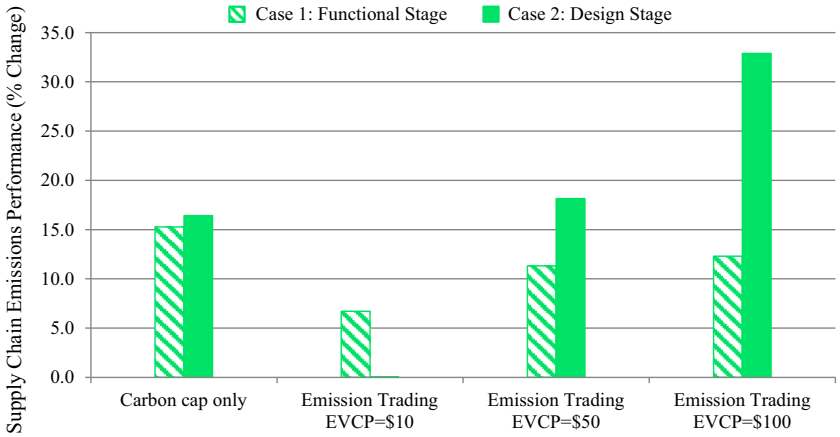


Fig. 9 Supply chain emissions performance at the functional and design stages under different carbon regulatory environments

5 Conclusions

This paper presented a two-stage stochastic model that can be used for design and sustainability analysis of a supply chain in a carbon trading environment. The uniqueness of the model lies in its ability to consider uncertainty in carbon price and product demand. When applied to a real world supply chain reconfiguration problem for an Australian manufacturer of office furniture, we found that: (1) pricing carbon emissions, particularly at higher prices, predominantly affects transportation operations and accordingly results in reduced transportation costs and emissions, (2) the higher the carbon price is, the greener would be the resulting supply chain configurations, not necessarily following a linear relationship, (3) in a carbon trading environment, the supply chain configuration can be highly sensitive to the probability distribution of the carbon credit price, (4) there is a nonlinear relationship between budget availability and supply chain greening; and (5) imposing an emissions cap

combined with a carbon credit price marginally increases total supply chain costs, but yields a greener design.

Our modeling effort and analysis come with limitations that can provide directions for further research in the area. Future research can consider larger supply chain networks and additional experimental scenarios that include resource use. Such investigations may more broadly evaluate efficient policy interventions for emissions reduction and controlled use of natural resources. Another direction for future research can be the development of multi-period supply chain planning models through multi-stage stochastic programming. Incorporating uncertainty in emissions rate of different transport modes or production systems can also be a critical area for future research. Finally, sophisticated heuristics solution methods may need to be designed to solve a greater number of scenarios within reasonable model runtimes.

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